

QUANTUM HACKS

TEAM NAME:-CODE KRAFTERS

Domain: Food Systems

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PROBLEM STATEMENT

6. FOOD DISTRIBUTION OPTIMIZATION FOR NGOS

Spoilage-Aware Quantum-Inspired Food Distribution System Tagline

Large quantities of food are wasted due to poor allocation. Relief organizations struggle with limited truck capacity. Perishable food often spoils before reaching needy people. Manual selection fails to maximize delivery impact.

1year – 80 million tonnes food wasted
Everyday – 190 million people stay hungry
Everymonth – 195 million people stay hungry



Why Existing Systems Fail

- Manual Planning: Prone to human error, slow, and doesn't scale with large datasets
- As food items increase, possible allocation combinations grow exponentially
- Manual entry means wrong expiry dates, quantities, or dietary restrictions → bad allocations.
- Traditional systems can't adapt to live inventory changes, sudden donations, or last-minute demand spikes
- Food safety laws need batch tracking. Manual systems can't prove where spoiled food came from..



SOLUTION

WE DESIGNED A:

QUANTUM FOOD SPOILAGE OPTIMIZER

FOOD DATA



PRIORITY SCORING



QFS OPTIMIZATION MODEL



BEST FOOD ALLOCATION

Quantum-Inspired Optimization QUBO-Based Optimization Engine

QUBO Mathematically Evaluates Multiple Food Combinations
And Helps Select The Most Optimal Allocation

QUBO FORMULATION
$$Q(X) = \sum I A_I X_I + \sum_{I,J} B_{IJ} X_I X_J$$

BINARY DECISION VARIABLES

- $X_I = 1 \rightarrow$ INCLUDE FOOD ITEM I IN ALLOCATION
- $X_I = 0 \rightarrow$ EXCLUDE FOOD ITEM I FROM ALLOCATION

QUBO = Quadratic Unconstrained Binary Optimization

Real-World Impact Impact & Applications

APPLICATIONS

NGO food distribution

Disaster relief logistics

Flood rescue operations

Smart ration allocation

Food bank optimization

BENIFITS

- ✓ Reduce food spoilage
- ✓ Improve truck utilization
- ✓ Deliver higher-priority food
- ✓ Faster allocation decisions
- ✓ Better humanitarian response



FUTURE SCOPE

FUTURE ENHANCEMENTS

1. AI-based demand prediction
2. Multi-truck optimization
3. Real-time traffic integration
4. IoT inventory monitoring
5. Cloud-based relief coordination
6. Integration with real quantum hardware Vision

QFSO

“Feeding Smarter With Quantum-Inspired Optimization”

FINAL STATEMENT:-

“we transformed food distribution into a spoilage-aware combinatorial optimization problem and solved it using qubo-based simulated annealing”.



Thank you

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